

ROOP SAGAR MANGINENI

Machine Learning Engineer | Software Engineer | AI Systems Builder

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EDUCATION

Master of Science in Computer Science, University of Illinois, USA	Jan 2024 – May 2025
Bachelor of Technology in Computer Science, SRM University, India	July 2019 – June 2023

PROFESSIONAL EXPERIENCE

Prosrvc LLC – USA, Software Engineer (Machine Learning)	May 2025 – Present
- Designed and deployed ML models for device resale price prediction and hardware defect detection, achieving 87%+ prediction accuracy and directly improving automated decision-making across all key internal business workflows significantly. - Built scalable data pipelines (Python, Pandas, SQL) processing 80,000+ device records monthly, increasing data reliability for ML model training, reducing downstream analytics errors, and improving overall data pipeline efficiency across all workflows. - REST APIs using FastAPI for real-time ML inference, integrating predictive models into systems with minimal latency. - Implemented model validation, logging, and monitoring frameworks improving reliability and observability of production ML services. - Collaborated with cross-functional teams to integrate ML models into backend applications and dashboards, improving efficiency.	
University of Illinois – USA, Graduate Research Assistant (Machine Learning / NLP)	Aug 2024 – Apr 2025
- Optimized BERT-based NLP models via advanced feature engineering and hyperparameter tuning, lifting text classification accuracy from 13% to 86.4%, a 6.6x improvement, directly contributing to a top-tier international peer-reviewed academic publication. - Designed large-scale NLP preprocessing pipelines in Python using tokenization and normalization techniques that reduced dataset noise and improved downstream model performance across multiple text classification tasks and real-world benchmarks. - Developed semantic similarity and document matching algorithms using cosine similarity, TF-IDF, and vector embeddings for fully automated text classification and scalable information retrieval across large, diverse multi-domain research datasets.	
Accelerize360 – USA, Machine Learning Engineer Intern	May 2024 – Aug 2024
- Developed and optimized ML models for enterprise applications using Python and Scikit-learn, significantly improving prediction accuracy and automating end-to-end workflows across multiple core business operations and internal production pipelines. - Built automated data preprocessing and feature engineering pipelines that eliminated manual data preparation bottlenecks, significantly improving ML workflow efficiency, reducing training time, and overall model training throughput. - Improved model performance through systematic feature selection, hyperparameter tuning, and structured cross-validation model evaluation, consistently delivering measurable accuracy improvements across all deployed production ML models.	

PROJECTS

Resumepedia – AI Resume Builder | Python, FastAPI, React, LLaMA 3.3 70B, Groq, PostgreSQL, Vercel, Render | [Link](#)

- **Architected a production-grade multi-agent AI resume builder** with a custom **5-agent orchestration system** (Writer, Critic, Tailor, Cover Letter, and Parser agents) running in parallel using Python ThreadPoolExecutor for fast concurrent generation.
- Built full-stack application: **React/Vite** frontend deployed on **Vercel**, **FastAPI** backend on **Render**, **PostgreSQL** database with user authentication, email verification, session management, and secure JWT-based token authorization.
- **Integrated Groq-hosted Llama 3.3 70B** via **REST API** for AI-powered resume generation, real-time critique with scoring rings, dynamic keyword tailoring to job descriptions, and automated job board integrations across **LinkedIn, Indeed, and Glassdoor** worldwide.
- Deployed live at resumepedia.vercel.app, accessible to any user worldwide with full signup, login, and resume management features

Job Search Web App | Python, Streamlit, NLP, Rapid API, Pandas | [Link](#)

- Built an **NLP-based job recommendation system** achieving **90%+** accuracy, matching profiles to listings via text similarity.
- Integrated CI/CD pipelines using Jenkins and GitHub Actions, scaling the platform to support **30%** higher concurrent user traffic.
- Implemented text similarity and cosine distance to improve job matching, reducing recommendations and improving accuracy

SKILLS

Languages: Python, SQL, JavaScript, C++
Frontend: React, Vite, Streamlit, HTML/CSS
Machine Learning: Scikit-learn, TensorFlow, PyTorch, NLP, LLMs, BERT, LangChain, Feature Engineering, Model Optimization
AI & Agents: Multi-Agent Systems, LLM Orchestration, Prompt Engineering, RAG, Groq, Llama 3.3 70B, Generative AI
Backend & Cloud: FastAPI, Flask, REST APIs, AWS (EC2, S3, Lambda), Docker, PostgreSQL, MySQL, Snowflake, CI/CD
Tools: Pandas, NumPy, Git, GitHub Actions, Linux, Vercel, Render, Retool

CERTIFICATIONS

Google Generative AI Leader Link	2026–2029
Covered LLM fundamentals, prompt engineering, AI governance, and scalable AI solutions using Google Cloud & Vertex AI.	

ACHIEVEMENTS

- 1st Prize — Hack SRM Hackathon:** Delivered an innovative, scalable solution that stood out among all competing teams.
- Served as **President, Data Science Club** at SRM AP — led workshops and ML initiatives mentoring **100+** peers in data science.